ClapCap: CLAP Prefix for Music Captioning

A frozen CLAP embedding → a short learned prefix → a small language model.

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Summary

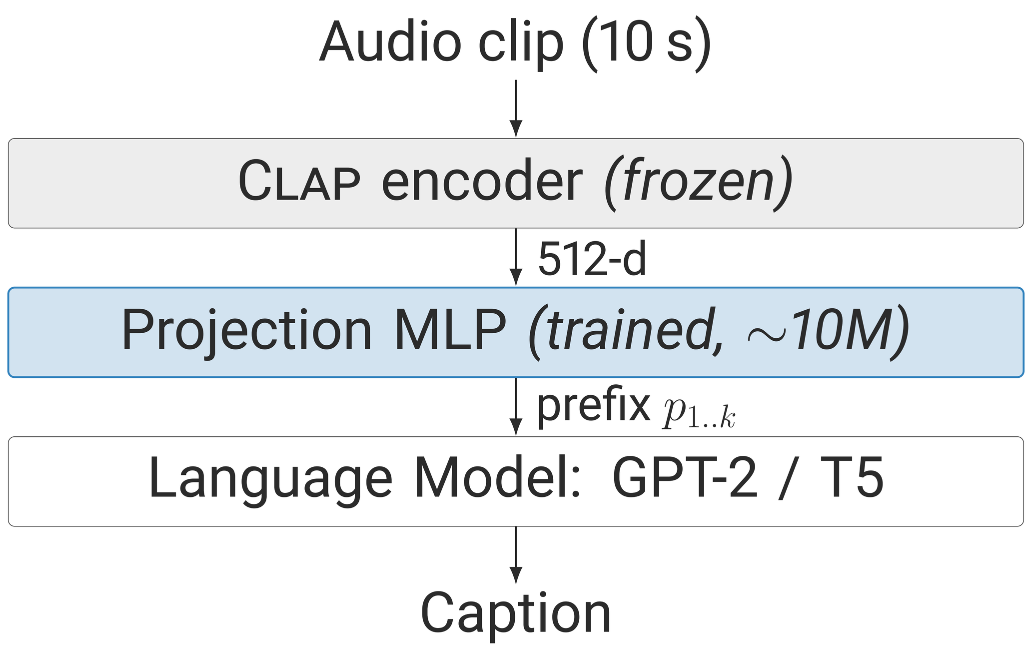
CLAPCAP is the audio analogue of the ClipCap image-captioning recipe: a **frozen** CLAP audio encoder turns a 10 s clip into one embedding, a small projection MLP maps it to a fixed-length prefix of pseudo-tokens, and a pretrained language model writes the caption. On MusicCaps we run a *controlled* comparison of two lightweight LM families — GPT-2 (124M) and T5-base (220M) — holding encoder, projection, data split and evaluation fixed, so the language model is the only variable.

One takeaway. A **124M** frozen-prefix LM captions music *competitively with* dedicated music captioners and clearly beats the clean ones on FENSE — one to two orders of magnitude smaller than billion-parameter audio LLMs, trained on a **single consumer GPU**.

Research question

Can a lightweight language model understand music semantics through a simple projection from a frozen audio embedding? And between two such models, *does the choice of LM matter* once run-to-run variance is accounted for?

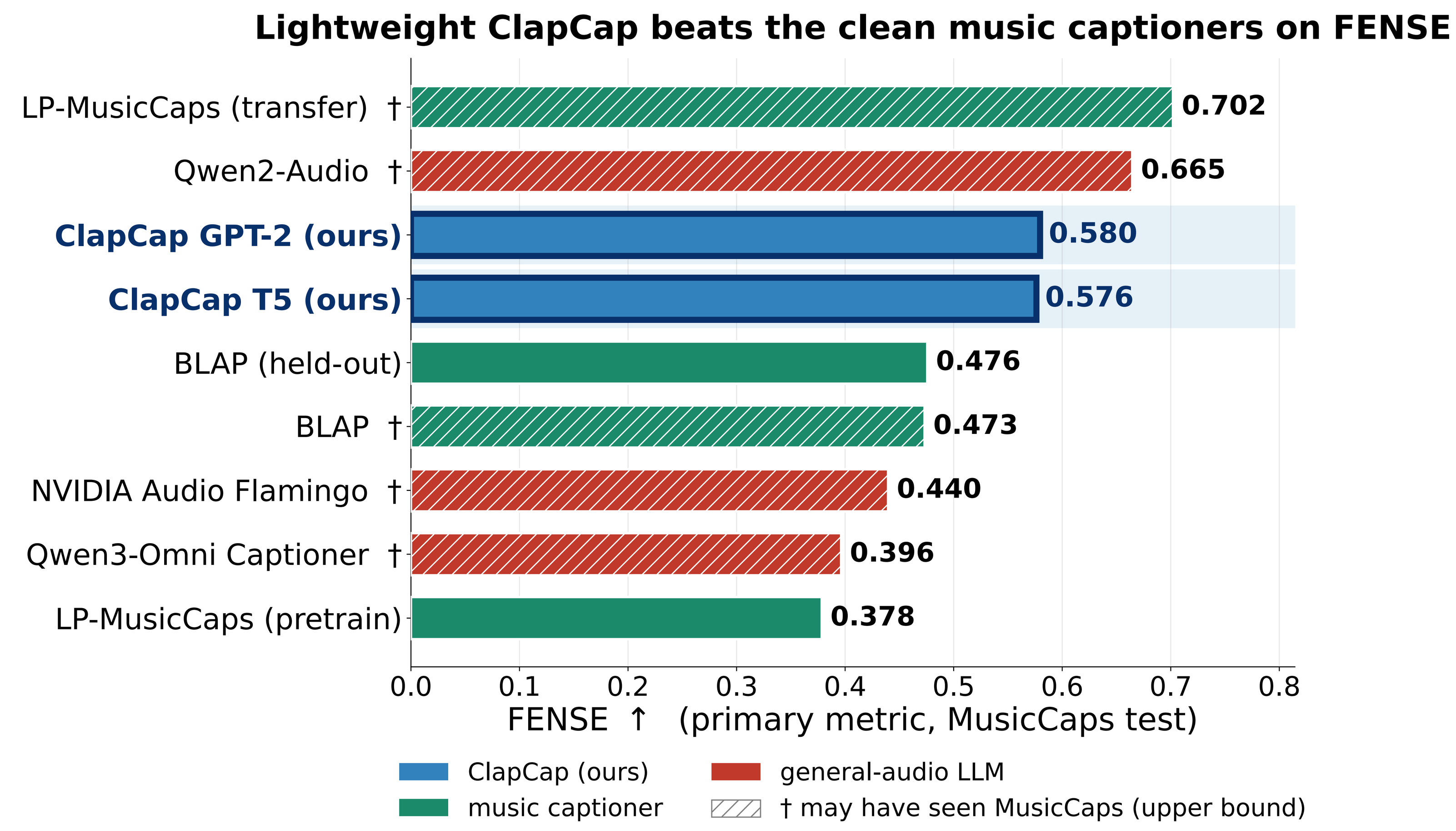
Method: the CLAPCAP pipeline



- **Only new module:** the projection MLP (~10.2M params); CLAP embeddings are precomputed once, so the encoder never runs in training.
- **Two-stage training:** Stage 1 trains the projection with the LM frozen; Stage 2 also fine-tunes the LM.
- **Evaluation:** a ten-metric suite with FENSE as the primary score; a per-model *noise floor* from three seeds, so every ablation delta is judged against run-to-run variance.

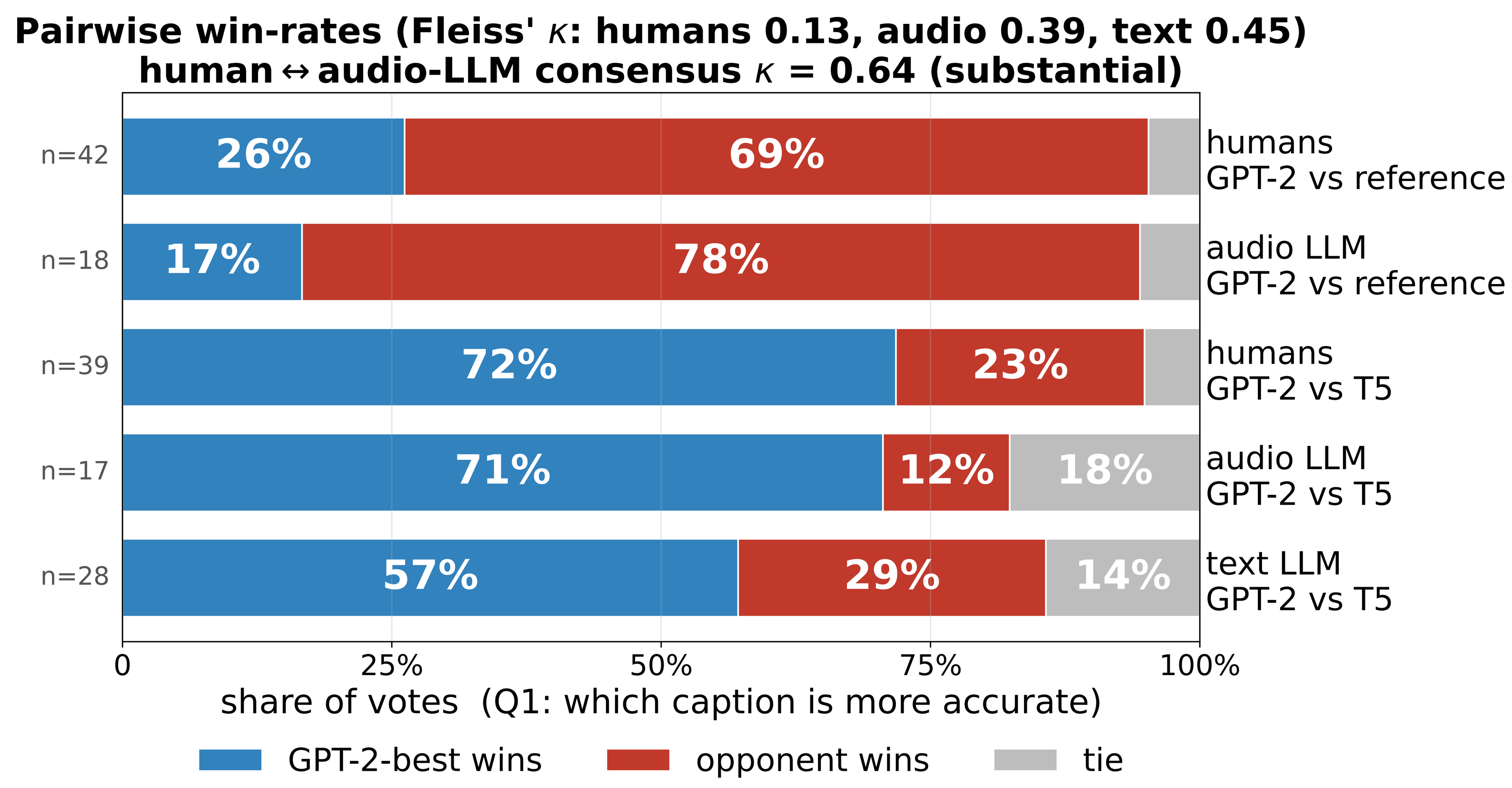
Result 1 — vs. reference captioners

On the shared MusicCaps test split the CLAPCAP best-decoding rows (GPT-2 FENSE **0.580**, T5 **0.576**) *outscore every clean reference captioner* — music-domain (LP-MusicCaps pretrain 0.378; BLAP 0.476, held-out) and general-audio (NVIDIA Audio Flamingo, Qwen2-Audio, Qwen3-Omni) alike. Rows marked † may have seen MusicCaps in training, so their bars are leakage upper bounds, not clean held-out scores.



Result 2 — pairwise A/B evaluation

A pairwise A/B study (more accurate? / less wrong?) across **three rater pools** — 10 human listeners, 3 audio LLM judges, 5 text LLM judges. Every pool **prefers GPT-2-best over T5-best**; the **reference still wins**, the honest ceiling. Lay humans are noisy individually (Fleiss κ 0.13) but their *consensus* tracks the audio panel (human↔LLM $\kappa = 0.64$).



Key findings

1. GPT-2 and T5 are **statistically indistinguishable at baseline** once the noise floor is accounted for.
2. Stage-2 fine-tuning **erases** nearly all architectural differences introduced in the projection stage.
3. The largest decoding lever (sentence completion) lifts FENSE mainly by **removing fluency errors**, not by adding semantic content.
4. Lightweight, single-GPU CLAPCAP is **competitive** with dedicated music captioners — the recipe, not scale, carries the result.

Conclusion

A frozen audio embedding plus a short learned prefix is enough for a 124M LM to describe music. The controlled analysis — and the efficiency of the recipe — is the contribution, not a single leaderboard number.

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